ROCSCIENCE

Case3
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:34:42 am
Software Version: 9.041

Table of Contents

Project Summary 3
General Settings 4
Analysis Options 5
Groundwater Analysis 6
Random Numbers 7
Surface Options 8
Seismic Loading 9
Materials 10
Global Minimums 11
Method: bishop simplified 11
Valid and Invalid Surfaces 12
Method: bishop simplified 12
Slice Data 13
Global Minimum Query (bishop simplified) - Safety Factor: 0.986442 13
Interslice Data 14
Global Minimum Query (bishop simplified) - Safety Factor: 0.986442 14
Entity Information 15
Group 1 15
Shared Entities 15

Slide2 Analysis Information

Case3

Project Summary

File Name:	Case3.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:03.127s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:34:42 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
Number of slices:	25
Tolerance:	0.0001
Maximum number of iterations:	100
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	20 kN/m3
Cohesion	3 kPa
Phi	19.6 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	0.986442
Center:	30.259, 51.792
Radius:	26.793
Left Slip Surface Endpoint:	30.000, 25.000
Right Slip Surface Endpoint:	51.137, 35.000
Resisting Moment:	10480.3 kN-m
Driving Moment:	10624.4 kN-m
Total Slice Area:	47.9165 m2
Surface Horizontal Width:	21.1374 m
Surface Average Height:	2.2669 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	9696
Number of Invalid Surfaces:	0

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.986442

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.845497	3.53064	0.35018	Material 1	3	19.6	4.53859	4.47706	4.14807	0	4.14807	4.17581	4.17581
2	0.845497	10.3661	2.15885	Material 1	3	19.6	7.36672	7.26684	11.9827	0	11.9827	12.2604	12.2604
3	0.845497	16.7492	3.96979	Material 1	3	19.6	9.94308	9.80827	19.1199	0	19.1199	19.8099	19.8099
4	0.845497	22.6775	5.78472	Material 1	3	19.6	12.2744	12.1079	25.5781	0	25.5781	26.8216	26.8216
5	0.845497	28.1475	7.6055	Material 1	3	19.6	14.3661	14.1713	31.3728	0	31.3728	33.291	33.291
6	0.845497	33.1538	9.43404	Material 1	3	19.6	16.223	16.003	36.5166	0	36.5166	39.2122	39.2122
7	0.845497	37.6897	11.2723	Material 1	3	19.6	17.8484	17.6064	41.0195	0	41.0195	44.577	44.577
8	0.845497	41.747	13.1225	Material 1	3	19.6	19.2452	18.9843	44.8893	0	44.8893	49.3757	49.3757
9	0.845497	45.3155	14.9867	Material 1	3	19.6	20.4155	20.1387	48.1311	0	48.1311	53.5963	53.5963
10	0.845497	48.383	16.8673	Material 1	3	19.6	21.3601	21.0705	50.7479	0	50.7479	57.2243	57.2243
11	0.845497	50.9352	18.7668	Material 1	3	19.6	22.0795	21.7801	52.7407	0	52.7407	60.2429	60.2429
12	0.845497	52.9553	20.6881	Material 1	3	19.6	22.573	22.267	54.108	0	54.108	62.6322	62.6322
13	0.845497	54.4237	22.6339	Material 1	3	19.6	22.8395	22.5298	54.846	0	54.846	64.369	64.369
14	0.845497	55.3176	24.6078	Material 1	3	19.6	22.8766	22.5664	54.9487	0	54.9487	65.4262	65.4262
15	0.845497	55.6103	26.6134	Material 1	3	19.6	22.6812	22.3737	54.4078	0	54.4078	65.7723	65.7723
16	0.845497	55.2706	28.6549	Material 1	3	19.6	22.2497	21.948	53.2121	0	53.2121	65.3706	65.3706
17	0.845497	54.262	30.737	Material 1	3	19.6	21.5766	21.2841	51.3476	0	51.3476	64.1777	64.1777
18	0.845497	52.5414	32.8652	Material 1	3	19.6	20.6561	20.376	48.7974	0	48.7974	62.1426	62.1426
19	0.845497	50.0575	35.0458	Material 1	3	19.6	19.4806	19.2165	45.5411	0	45.5411	59.2048	59.2048
20	0.845497	46.7489	37.2864	Material 1	3	19.6	18.0415	17.7969	41.5545	0	41.5545	55.2917	55.2917
21	0.845497	42.5414	39.5961	Material 1	3	19.6	16.3285	16.1071	36.8091	0	36.8091	50.3153	50.3153
22	0.845497	37.3435	41.9856	Material 1	3	19.6	14.3296	14.1353	31.2716	0	31.2716	44.1675	44.1675
23	0.845497	31.0414	44.4688	Material 1	3	19.6	12.031	11.8679	24.9039	0	24.9039	36.7139	36.7139
24	0.845497	23.0637	47.063	Material 1	3	19.6	9.28569	9.15979	17.2987	0	17.2987	27.2783	27.2783
25	0.845497	8.4565	49.7907	Material 1	3	19.6	4.66124	4.59804	4.48781	0	4.48781	10.0018	10.0018

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.986442

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30	25	0	0	0
2	30.8455	25.0052	3.81592	0	0
3	31.691	25.037	9.66252	0	0
4	32.5365	25.0957	16.9475	0	0
5	33.382	25.1814	25.1345	0	0
6	34.2275	25.2943	33.7391	0	0
7	35.073	25.4348	42.3254	0	0
8	35.9185	25.6033	50.5034	0	0
9	36.764	25.8004	57.9273	0	0
10	37.6095	26.0267	64.2945	0	0
11	38.455	26.2831	69.3449	0	0
12	39.3005	26.5704	72.8614	0	0
13	40.146	26.8896	74.6709	0	0
14	40.9915	27.2422	74.6464	0	0
15	41.837	27.6294	72.7101	0	0
16	42.6825	28.0531	68.8376	0	0
17	43.528	28.5151	63.0639	0	0
18	44.3735	29.0178	55.4913	0	0
19	45.2189	29.5641	46.3004	0	0
20	46.0644	30.1571	35.7638	0	0
21	46.9099	30.8009	24.2659	0	0
22	47.7554	31.5003	12.3288	0	0
23	48.6009	32.2612	0.64966	0	0
24	49.4464	33.0911	-9.84755	0	0
25	50.2919	33.9998	-17.7156	0	0
26	51.1374	35	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	70, 20 70, 35 50, 35 30, 25 20, 25 20, 20